

Probability
Chapter 3 - Module 1

Introduction to Conditioning

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Recall that for events E and F , we have

$$P(E|F) = \frac{P(EF)}{P(F)}.$$

This extends easily to the discrete case in which

$$P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}$$

or equivalently

$$p_{X|Y}(x|y) = \frac{p(x, y)}{p_Y(y)}$$

as long as

$$p_Y(y) = P(Y = y) \neq 0.$$

You can not condition on an event that has probability zero.

Def. (Conditional Mass Function) For discrete random variables, X and Y , for $p_Y(y) \neq 0$, define the conditional mass function as

$$p_{X|Y}(x|y) = \frac{p(x, y)}{p_Y(y)}.$$

Note that this conditional mass function is a probability mass function in x for each y . (You should verify this.)

Def. (Conditional CDF) Let X and Y be discrete random variables.

If $p_Y(y) \neq 0$, define

$$F_{X|Y}(x|y) = P(X \leq x | Y = y) = \sum_{a \leq x} p_{X|Y}(a|y).$$

Def. (Conditional Expectation) Let X and Y be discrete random variables.

If $p_Y(y) \neq 0$, define

$$E_{X|Y}[x|y] = \sum_x x p_{X|Y}(x|y).$$

Example 1. Let

$$\begin{array}{lll} p(1, 1) = 1/10 & p(1, 2) = 1/5 & p(1, 3) = 1/20 \\ p(2, 1) = 0 & p(2, 2) = 1/20 & p(2, 3) = 2/10 \\ p(3, 1) = 2/10 & p(3, 2) = 1/10 & p(3, 3) = 1/10 \end{array}$$

Then

$$\begin{array}{lll} p_X(1) = 7/20 & p_X(2) = 5/20 & p_X(3) = 8/20 \\ p_Y(1) = 6/20 & p_Y(2) = 7/20 & p_Y(3) = 7/20. \end{array}$$

So

$$p_{X|Y}(1|2) = \frac{p(1, 2)}{p_Y(2)} = \frac{1/5}{7/20} = \frac{20}{35} = \frac{4}{7}.$$

$$\begin{aligned}
F_{X|Y}(2|3) &= P(X \leq 2|Y = 3) \\
&= P(X = 1|Y = 3) + P(X = 2|Y = 3) \\
&= \frac{p(1, 3)}{p_Y(3)} + \frac{p(2, 3)}{p_Y(3)} \\
&= \frac{1/20}{7/20} + \frac{2/10}{7/20} \\
&= \frac{1}{7} + \frac{4}{7} = \frac{5}{7}.
\end{aligned}$$

And finally

$$\begin{aligned}E_{X|Y}[X|Y = 3] &= \sum_x x P(X = x|Y = 3) \\&= \sum_x x \frac{p(x, 3)}{p_Y(3)} \\&= 1 \frac{p(1, 3)}{p_Y(3)} + 2 \frac{p(2, 3)}{p_Y(3)} + 3 \frac{p(3, 3)}{p_Y(3)} \\&= 1 \frac{1/20}{7/20} + 2 \frac{2/10}{7/20} + 3 \frac{1/10}{7/20} \\&= 1 \frac{1}{7} + 2 \frac{4}{7} + 3 \frac{2}{7} \\&= \frac{15}{7}.\end{aligned}$$

Example 2.

3.3 from the textbook

Let X and Y be independent binomial random variables with respective parameters (n_1, p) and (n_2, p) . Calculate the conditional expected value of X given that $X + Y = m$.

Solution. With $q = (1 - p)$

$$\begin{aligned} P(X = k | X + Y = m) &= \frac{P(X = k, X + Y = m)}{P(X + Y = m)} \\ &= \frac{P(X = k, Y = m - k)}{P(X + Y = m)} \\ &= \frac{P(X = k)P(Y = m - k)}{P(X + Y = m)} \\ &= \frac{\binom{n_1}{k} p^k q^{n_1 - k} \binom{n_2}{m - k} p^{m - k} q^{n_2 - (m - k)}}{\binom{n_1 + n_2}{m} p^m q^{n_1 + n_2 - m}} \\ &= \frac{\binom{n_1}{k} \binom{n_2}{m - k}}{\binom{n_1 + n_2}{m}}. \end{aligned}$$

Comments.

Since X is a binomial random variable with parameters (n_1, p) ,

$$P(X = k) = \binom{n_1}{k} p^k q^{n_1-k}.$$

Since Y is a binomial random variable with parameters (n_2, p) ,

$$P(Y = m - k) = \binom{n_2}{m - k} p^{m-k} q^{n_2-(m-k)}.$$

And finally we use the fact that $X + Y$ is a binomial random variable with parameters $(n_1 + n_2, p)$ to write

$$P(X + Y = m) = \binom{n_1 + n_2}{m} p^m q^{n_1+n_2-m}.$$

The final solution was

$$P(X = k | X + Y = m) = \frac{\binom{n_1}{k} \binom{n_2}{m-k}}{\binom{n_1+n_2}{m}}$$

We have not come across this formula before, but this is the distribution for a random variable W with a hypergeometric density.

This discrete density comes from an experiment where we determine the probability of choosing m white balls from an urn that contains n_1 white balls and n_2 black balls.

The mean of this random variable is

$$m \frac{n_1}{n_1 + n_2}.$$

We conclude that

$$E[X | X + Y = m] = m \frac{n_1}{n_1 + n_2}.$$

We note with some surprise that the answer is independent of p . This is an effect of the conditioning.

Fact. Let X be a binominal random variable with parameters (n_1, p) and let Y be a binominal random variable with parameters (n_2, p) . If X and Y are independent, then $X + Y$ is a binomial random variable with parameters $(n_1 + n_2, p)$

This follow from the fact that moment generating functions for X and Y are

$$M_1(t) = (pe^t + (1 - p))^{n_1}$$

and

$$M_2(t) = (pe^t + (1 - p))^{n_2}.$$

Since X and Y are independent, then the moment generating function for $X + Y$ is

$$\begin{aligned} M(t) &= M_1(t)M_2(t) \\ &= (pe^t + (1 - p))^{n_1} (pe^t + (1 - p))^{n_2} \\ &= (pe^t + (1 - p))^{n_1 + n_2} \end{aligned}$$

This makes sense because X models the number of successes with n_1 trials with probability of success p and Y models the number of successes with n_2 trials with probability of success p . So naturally, $X + Y$ models the number of success in $n_1 + n_2$ trials with probability of success p .

Def. (Continuous Conditional Density Function) Let X and Y be continuous random variables with joint pdf $f(x, y)$. Given that $f_Y(y) > 0$ we define

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}.$$

Motivation.

$$\begin{aligned} P(X \leq x | y \leq Y \leq y + \Delta y) &= \frac{P(X \leq x \wedge y \leq Y \leq y + \Delta y)}{P(y \leq Y \leq y + \Delta y)} \\ &= \frac{\frac{1}{\Delta y} \int_{-\infty}^x \int_y^{y+\Delta y} f(u, v) dv du}{\frac{1}{\Delta y} \int_y^{y+\Delta y} f_Y(v) dv} \\ &= \frac{\int_{-\infty}^x \left(\frac{1}{\Delta y} \int_y^{y+\Delta y} f(u, v) dv \right) du}{\frac{1}{\Delta y} \int_y^{y+\Delta y} f_Y(v) dv} \\ &\rightarrow \frac{\int_{-\infty}^x f(u, y) du}{f_Y(y)} \end{aligned}$$

under the assumption that $f(u, v)$ is continuous at (u, y) for all values of u and that $f_Y(v)$ is continuous at $y = v$.

This makes sense?

Recall that for continuous random variables,

$$P(X \leq x) = \int_{-\infty}^x f(u) du.$$

And at points of continuity of f ,

$$\frac{d}{dx} P(X \leq x) = \frac{d}{dx} \int_{-\infty}^x f(u) du = f(x)$$

Similarly

$$\frac{d}{dx} \frac{\int_{-\infty}^x f(u, y) du}{f_Y(y)} = \frac{\frac{d}{dx} \int_{-\infty}^x f(u, y) du}{f_Y(y)} = \frac{f(x, y)}{f_Y(y)}.$$

Example 3. Let $(X, Y) \sim N(\mu, \Sigma)$ in two dimensions with density

$$f(x, y) = \frac{1}{(\sqrt{2\pi})^2} |\Sigma|^{-1/2} \text{Exp} \left(-(x - \mu_x, y - \mu_y) \Sigma^{-1} \begin{pmatrix} x - \mu_x \\ y - \mu_y \end{pmatrix} / 2 \right),$$

where

$$\Sigma = \begin{pmatrix} \sigma_x^2 & \rho \sigma_x \sigma_y \\ \rho \sigma_x \sigma_y & \sigma_y^2 \end{pmatrix}$$

so that

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-(y - \mu_y)^2 / 2\sigma_y^2}.$$

Then

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$$

Let's look at some examples in Mathematica.

Problem. Let (X, Y) have pdf $f(x, y)$. Let $T = X + Y$. Determine $f_{X|T}(x | t)$.

Solution. Consider the change of variables

$$x = g_1(x, y) = x \quad \text{and} \quad t = g_2(x, y) = x + y.$$

Then the inverse mapping is

$$x = h_1(x, t) = x \quad \text{and} \quad y = h_2(x, t) = t - x.$$

In matrix form this becomes the transformation is

$$\begin{pmatrix} x \\ t \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

It is easy to see that the Jacobian is

$$J(x, y) = \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = 1$$

and so $|J(x, y)| = 1$. Therefore

$$f_{X, X+Y}(x, t) = f(x, y) \Big|_{y=t-x} = f(x, t - x).$$

Problem. Let X_1 and X_2 be independent exponential densities. Determine $f_{X_1|X_1+X_2}(x|t)$.

Solution. Since X_1 and X_2 are exponential densities, let

$$X_1 \sim \mu_1 e^{-\mu_1 x_1} \quad \text{for } x_1 \geq 0$$

$$X_2 \sim \mu_2 e^{-\mu_2 x_2} \quad \text{for } x_2 \geq 0.$$

Since X_1 and X_2 are independent, then

$$f(x_1, x_2) = \begin{cases} \mu_1 e^{-\mu_1 x_1} \mu_2 e^{-\mu_2 x_2} & \text{for } x_1, x_2 \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

And from the previous result,

$$\begin{aligned} f_{X_1, X_1+X_2}(x, t) &= \begin{cases} \mu_1 e^{-\mu_1 x} \mu_2 e^{-\mu_2 (t-x)} & \text{for } x, t-x \geq 0 \\ 0 & \text{otherwise} \end{cases} \\ &= \begin{cases} \mu_1 e^{-\mu_1 x} \mu_2 e^{-\mu_2 (t-x)} & \text{for } 0 \leq x \leq t \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

So

$$f_{X_1, X_1+X_2}(x, t) = \begin{cases} \mu_1 e^{-\mu_1 x} \mu_2 e^{-\mu_2(t-x)} & \text{for } 0 \leq x \leq t \\ 0 & \text{otherwise.} \end{cases}$$

and

$$\begin{aligned} f_{X_1|X_1+X_2}(x|t) &= \frac{f_{X_1, X_1+X_2}(x, t)}{f_{X_1+X_2}(t)} \\ &= \frac{\mu_1 e^{-\mu_1 x} \mu_2 e^{-\mu_2(t-x)}}{f_{X_1+X_2}(t)} \\ &= C_t e^{-(\mu_1 - \mu_2)x}, \end{aligned}$$

for $0 \leq x \leq t$ and where

$$C_t = \frac{\mu_1 \mu_2 e^{-\mu_2 t}}{f_{X_1+X_2}(t)}.$$

$$f_{X_1|X_1+X_2}(x|t) = C_t e^{-(\mu_1 - \mu_2)x} \quad \text{for } 0 \leq x \leq t$$

Case 1. $\mu_1 = \mu_2$.

Then

$$f_{X_1|X_1+X_2}(x|t) = C_t,$$

for $0 \leq x \leq t$ and zero elsewhere. But for each t , $f_{X_1|X_1+X_2}(x|t)$ is a density function in x and integrates to 1. Therefore $C_t = \frac{1}{t}$ and as a biproduct we get that

$$f_{X_1+X_2} = \frac{\mu_1 \mu_2 e^{-\mu_2 t}}{C_t} = \mu_1 \mu_2 t e^{-\mu_2 t}.$$

$$f_{X_1, X_1+X_2}(x, t) = \mu_1 e^{-\mu_1 x} \mu_2 e^{-\mu_2(t-x)} \quad \text{for } 0 \leq x \leq t$$

Case 2. $\mu_1 \neq \mu_2$

We could compute this in the same way, but instead, let's compute $f_{X_1+X_2}$ as a marginal density function for $f_{X_1, X_1+X_2}(x, t)$ by integrating out x .

So

$$\begin{aligned} f_{X_1+X_2}(t) &= \int_0^t \mu_1 e^{-\mu_1 x} \mu_2 e^{-\mu_2(t-x)} dx \\ &= \int_0^t \mu_1 \mu_2 e^{-\mu_2 t} e^{-(\mu_1 - \mu_2)x} dx \\ &= \mu_1 \mu_2 e^{-\mu_2 t} \int_0^t e^{-(\mu_1 - \mu_2)x} dx \\ &= \mu_1 \mu_2 e^{-\mu_2 t} \frac{e^{-(\mu_1 - \mu_2)x}}{\mu_2 - \mu_1} \Big|_0^t \\ &= \frac{\mu_1 \mu_2}{\mu_2 - \mu_1} e^{-\mu_2 t} \left(1 - e^{(\mu_1 - \mu_2)t} \right) \\ &= \frac{\mu_1 \mu_2}{\mu_2 - \mu_1} \left(e^{-\mu_1 t} - e^{-\mu_2 t} \right). \end{aligned}$$

As a final check on our calculations, we note that

$$\begin{aligned}\lim_{\mu_1, \mu_2 \rightarrow \mu} \frac{\mu_1 \mu_2}{\mu_2 - \mu_1} \left(e^{-\mu_1 t} - e^{\mu_2 t} \right) &= \lim_{\mu_1, \mu_2 \rightarrow \mu} \mu_1 \mu_2 \frac{\left(e^{-\mu_1 t} - e^{\mu_2 t} \right)}{\mu_2 - \mu_1} \\&= \lim_{\mu_1, \mu_2 \rightarrow \mu} -\mu_1 \mu_2 \frac{\left(e^{-\mu_1 t} - e^{\mu_2 t} \right)}{\mu_1 - \mu_2} \\&= -\mu^2 \frac{d}{du} e^{-\mu t} \\&= \mu^2 t e^{-\mu t},\end{aligned}$$

which agrees with our earlier calculation.

So in conclusion, when $\mu_1 \neq \mu_2$,

$$\begin{aligned} f_{X_1|X_1+X_2}(x | t) &= \frac{\mu_1 e^{-\mu_1 x} \mu_2 e^{-\mu_2(t-x)}}{\frac{\mu_1 \mu_2}{\mu_2 - \mu_1} (e^{-\mu_1 t} - e^{-\mu_2 t})} \\ &= \frac{(\mu_1 - \mu_2) e^{-(\mu_1 - \mu_2)x}}{1 - e^{-(\mu_1 - \mu_2)t}}. \end{aligned}$$